14

Sequences and Series

Paper 1

* Remember the proofs of the sum formulae.

QUESTION 1

1.1 The following arithmetic sequence is given: 20 ; 23 ; 26 ; 29; … ; 101

1.1.1 How many terms are there in this sequence? (2)

1.1.2 The even numbers are removed from the sequence.
Calculate the sum of the terms of the remaining sequence. (6)

1.2 Study the geometric series: $x + \frac{x^2}{3} + \frac{x^3}{9} + \cdots$ ....

1.2.1 Determine the n-th term in terms of x. (2)

1.2.2 Determine the value(s) of x for which the series will converge. (3)

1.3 The sum of n terms in a sequence is given by $S_n = -n^2 + 5$. Determine the 23rd term. (3)

[20]

QUESTION 2

The sequence 3 ; 9 ; 17 ; 27 ; … is quadratic.

2.1 Determine an expression for the n-th term of the sequence. (4)

2.2 What is the value of the first term of the sequence that is greater than 269? (4)

QUESTION 3

3.1 Given the following number pattern which is a combination of a linear and a geometric pattern:

$$3 \ ; \ \frac{1}{2} \ ; \ 3 \ ; \ \frac{4}{10} \ ; \ 3 \ ; \ \frac{16}{50} \ ; \ \ldots$$

3.1.1 Write down the values of the next TWO terms of the pattern. (2)

3.1.2 Calculate the sum of the first thirty-five terms of the pattern. (5)

3.2 Calculate:

$$\sum_{n=3}^{\infty} 5.3^{1-n}$$

(4)

[11]

QUESTION 4

4.1 Given the sequence: $4 \; ; \; x \; ; \; 32$

Determine the value(s) of x if the sequence is:

4.1.1 Arithmetic (2)

4.1.2 Geometric (3)

4.2 Determine the value of P if $P = \sum_{k=1}^{13} 3^{k-5}$ (4)

QUESTION 5

A quadratic pattern has a second term equal to 1, a third term equal to – 6 and a fifth term equal to – 14.

5.1 Calculate the second difference of this quadratic pattern. (5)

5.2 Hence, or otherwise, calculate the first term of the pattern. (2)

QUESTION 6

6.1. A ball bounces $\frac{1}{2}$ of its previous height on each bounce. Find the total distance covered by the ball, until it comes to rest, if it is dropped from a height of 20 metres.

6.2. A specific tree grows 1,5 m in the 1st year. Its growth each year thereafter is $\frac{2}{3}$ of its growth in the previous year. What is the greatest height it can reach?

QUESTION 7

7.1 Given the quadratic sequence: $3\ ;\ -1\ ;\ 2\ ;\ \ldots$

7.1.1 Write down the fourth term of the sequence. (1)

7.1.2 Determine a formula for the n-th term of the sequence. (4)

7.2 Determine the value of the greatest term of the sequence defined by

$T_n = 1000 - \left(n - \dfrac{151}{4}\right)^2$. *Give the exact value, in decimal form.* (3)

[8]

QUESTION 8

Let $S = \sum_{p=1}^{n} (3p - 5)$.

8.1 If $n = 100$, calculate the value of S. (4)

8.2 If $S = 530$, calculate the value of n. (6)

[10]

8.3 Given the series $54 + 18 + 6 + \ldots$

8.3.1 Determine the nth term of this series. (2)

8.3.2 Hence determine the 12^{th} term of the series. (2)

8.3.3 Show that the sum to n terms of this series is $81 - 81(\frac{1}{3})^n$. (3)

8.3.4 Calculate: $\sum_{n=0}^{10} 54\left(\dfrac{1}{3}\right)^n$ (3)

8.3.5 Determine the maximum value of: $\sum_{n=0}^{k} 54\left(\dfrac{1}{3}\right)^n$ (3)

QUESTION 9

9.1 Determine the formula for the n-th term of the quadratic sequence:

$$-1\ ;\ 2\ ;\ 10\ ;\ 23\ ;\ 41\ ;\ ...$$ (4)

9.2 Consider the series: $a^{100}+a^{97}+a^{94}+a^{91}+...+a^{13}$

9.2.1 Write the series in sigma notation. (5)

9.2.2 Determine the sum of the series, to the nearest whole number, if $a=1{,}1$. (5)

9.3 In an arithmetic sequence, the 21^{st} term is 7 and the sum of the first 10 terms is -550. Determine the first term of the sequence. (6)

9.4 Consider the series: $\sum_{k=0}^{\infty}\left(x^2-4\right)\left(x+3\right)^k$

9.4.1 For which values of x does the series converge? (3)

9.4.2 Assuming that x is a value for which the series converges, show that

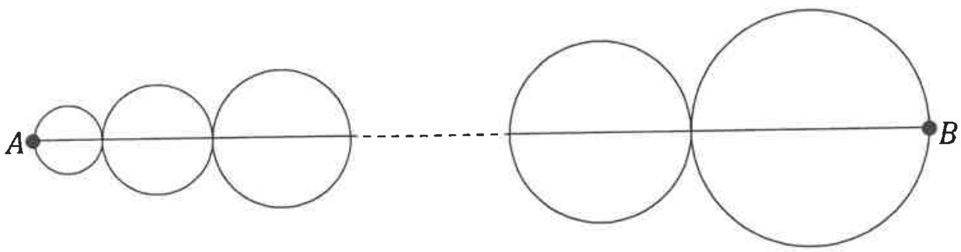
$$\sum_{k=0}^{\infty}\left(x^2-4\right)\left(x+3\right)^k=2-x.$$ (4)

QUESTION 10

10.1 Calculate the value of p if $\sum_{k=1}^{\infty}27p^k=\sum_{t=1}^{12}(24-3t)$. (5)

10.2 For which values of x will the geometric series $\sum_{k=1}^{\infty}4(3-x)^k$ converge? (4)

10.3 A number of circles touch each other as shown below:



The area of the smallest circle is 4π cm^2 and each consecutive circle has an area of $\frac{9}{4}$ times that of the previous one. If the distance of $AB=\frac{665}{8}$ cm, how many circles are there? (5)

QUESTION 11

11.1 If $\dfrac{a+b}{a-b} = \dfrac{b+c}{b-c}$, show that a, b and c form a geometric sequence. (3)

11.2 In an arithmetic progression the eighth term is twice the fourth term. The sum of the first three terms is 12. Determine the first term. (5)

11.3 The sum of n terms of an arithmetic series is $5n^2 - 11n$, for all values of n. Determine the common difference. (5)

11.4 11.4.1 For which values of k does the series $\sum_{n=1}^{\infty} (k - \frac{3}{2})^n$ converge? (3)

11.4.2 If the series does converge, prove: $\sum_{n=1}^{\infty} (k - \frac{3}{2})^n = \dfrac{2k-3}{5-2k}$ (4)

11.4.3 Hence, or otherwise, evaluate $\sum_{n=1}^{\infty} (k - \frac{3}{2})^n$, if possible:

11.4.3 (a) when $k = 1$

11.4.3 (b) when $k = -1$ (3)

11.5 A hot-dog vendor stores his supplies at his house. On his first day of business, he loads his trolley and pushes it the length of a block which is 79 metres away from his house. At sunset he returns home. The next day he walks double the distance, 158 metres, before he starts to sell his goods. On the third day he walks 3 blocks, which is 237 metres from his house. Every day he walks one block further. If all the blocks are of the same length, calculate how many kilometres he will walk in 28 days. (Leave your answer correct to the nearest kilometre.)

(6)

Sequences and Series : Solutions

Question 1

1.1.1. $T_n = a + (n-1)d$

$101 = 20 + (n-1)(3)$

$27 = n - 1$

$\underline{28 = n}$

1.1.2. Sequence: $23 ; 29 ; 35 ; \ldots ; 101$

$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$

$= \frac{14}{2}\left[2(23) + (14-1)(6)\right]$

$= 7\left[46 + 78\right]$

$= \underline{868}$

1.2.1. $r = \frac{x^2}{3} \div x$

$= \underline{x/3}$

$T_n = ar^{n-1}$

$= \underline{x\left(x/3\right)^{n-1}}$

1.2.2. $-1 < x/3 < 1$

$\underline{-3 < x < 3}$

1.3 $T_{23} = S_{23} - S_{22}$

$= -(23)^2 + 5 - \left[-(22)^2 + 5\right]$

$= -524 + 479$

$= \underline{-45}$

Question 2

2.1.

3   9   17   27

6   8   10

2   2

$2a = 2$

$\underline{a = 1}$

$3a + b = 6$

$3(1) + b = 6$

$\underline{b = 3}$

$a + b + c = 3$

$1 + 3 + c = 3$

$\underline{c = -1}$

$\underline{T_n = n^2 + 3n - 1}$

2.2.

$n^2 + 3n - 1 > 269$

$n^2 + 3n - 270 > 0$

$(n + 18)(n - 15) > 0$

$n < -18$ OR $n > 15$

N.A

∴ 16th term will be bigger than 269.

Question 3:

3.1.1. $3 \quad ; \quad {}^{32}/_{125}$

3.1.2. $18 \times 3 + \dfrac{\frac{1}{2}\left[1 - \left({}^{4}/_{5}\right)^{17}\right]}{1 - {}^{4}/_{5}}$

$= 54 + 2{,}44$

$= \underline{56{,}44}$

3.2. $\sum_{n=3}^{\infty} 5.3^{1-n}$

$= 5.3^{1-3} + 5.3^{1-4} + 5.3^{1-5} + \ldots$

$= \dfrac{5}{9} + \dfrac{5}{27} + \dfrac{5}{81} + \ldots$

$S_\infty = \dfrac{a}{1-r}$

$= \dfrac{{}^{5}/_{9}}{1 - {}^{1}/_{3}}$

$= \underline{{}^{5}/_{6}}$

Question 4:

4.1.1. $x - 4 = 32 - x$

$2x = 36$

$\underline{x = 18}$

4.1.2. $\dfrac{x}{4} = \dfrac{32}{x}$

$x^2 = 128$

$\underline{x = \pm 8\sqrt{2}}$

4.2. $\sum_{k=1}^{13} 3^{k-5}$

$= \frac{1}{81} + \frac{1}{27} + \frac{1}{9} + \ldots$

$S_{13} = \frac{\frac{1}{81}\left[(3)^{13} - 1\right]}{3-1}$

$= \frac{797161}{81}$

Question 5

5.1. $T_3 - T_2 = -6 - 1$

$9a + 3b + c - 4a - 2b - c = -7$

$5a + b = -7$

$T_5 = 25a + 5b + c = -14$

$5(5a+b) + c = -14$

$5(-7) + c = -14$

$c = 21$

T_2: $4a + 2b + 21 = 1$

$4a + 2b = -20$

$2a + b = -10 \ldots (1)$

T_3: $9a + 3b + 21 = -6$

$9a + 3b = -27$

$3a + b = -9 \ldots (2)$

$(2) - (1)$ $a = 1$

$\therefore 2a = 2$

5.2. $2(1) + b = -16$

$\underline{b = -12}$

$T_1 = a + b + c$

$= 1 - 12 + 21$

$= \underline{10}$

Question 6

6.1. $20 + 2\left[\dfrac{10}{1-\frac{1}{2}}\right]$

$= \underline{60\,m}$

6.2. $S_\infty = \dfrac{a}{1-r}$

$= \dfrac{1{,}5}{1-\frac{2}{3}}$

$= \underline{4{,}5\,m}$

Question 7

7.1.1. $\underline{12}$

7.1.2.

$3 \quad -1 \quad 2$

$-4 \quad 3$

7

$2a = 7$

$\underline{a = \frac{7}{2}}$

$3a + b = -4$

$3\left(\frac{7}{2}\right) + b = -4$

$\underline{b = \frac{-29}{2}}$

$a + b + c = 3$

$\frac{7}{2} - \frac{29}{2} + c = 3$

$-11 + c = 3$

$c = 14$

$T_n = \frac{7}{2}n^2 - \frac{29}{2}n + 14$

7.2. $T_{38} = 1000 - \left(38 - \frac{151}{4}\right)^2$

$= 1000 - \frac{1}{16}$

$= 999{,}9375$

Question 8:

8.1. $\sum_{p=1}^{100} (3p - 5)$

$= -2 + 1 + 4 + 7 + \ldots$

$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$

$= \frac{100}{2}\left[2(-2) + 99(3)\right]$

$= 14\,650$

8.2 $\frac{n}{2}\left[2(-2) + (n-1)3\right] = 530$

$n(3n - 7) = 1060$

$3n^2 - 7n - 1060 = 0$

$n = \frac{7 \pm \sqrt{49 - 4(3)(-1060)}}{2(3)}$

$n = \frac{7 \pm 113}{6}$

$n = 20$ OR $n = \frac{-53}{3}$ N.A

8.3.1. $54 + 18 + 6 + \ldots$

$T_n = ar^{n-1}$
$= 54\left(\frac{1}{3}\right)^{n-1}$

8.3.2 $T_{12} = 54\left(\frac{1}{3}\right)^{12-1}$
$= \frac{2}{6561}$

8.3.3. $S_n = \frac{a[1 - r^n]}{1 - r}$
$= \frac{54\left[1 - \left(\frac{1}{3}\right)^n\right]}{1 - \frac{1}{3}}$
$= \frac{54\left[1 - \left(\frac{1}{3}\right)^n\right]}{2/3}$
$= 81\left(1 - \left(\frac{1}{3}\right)^n\right)$
$= 81 - 81\left(\frac{1}{3}\right)^n$

8.3.4. $S_{11} = 81 - 81\left(\frac{1}{3}\right)^{11}$
$= 80,99954275$
≈ 81

8.3.5 $S_\infty = \frac{54}{1 - \frac{1}{3}}$
$= 81$

Question 9:

9.1. -1 2 10 23 41

3 8 13 18

5 5 5

$2a = 5$

$a = \frac{5}{2}$

$3a + b = 3$

$3(\frac{5}{2}) + b = 3$

$b = -\frac{9}{2}$

$a + b + c = -1$

$\frac{5}{2} - \frac{9}{2} + c = -1$

$-2 + c = -1$

$c = 1$

$T_n = \frac{5}{2}n^2 - \frac{9}{2}n + 1$

9.2.1. $n = \left(\frac{100 - 13}{3}\right) + 1$

$= 30$

$r = \frac{a^{97}}{a^{100}} = \frac{1}{a^3}$

$$\sum_{k=1}^{30} a^{100} \left(\frac{1}{a^3}\right)^{k-1}$$

9.2.2. $S_n = \frac{a(1 - r^n)}{1 - r}$

$= \frac{(1,1)^{100}\left(1 - \left(\frac{1}{1,1^3}\right)^{30}\right)}{1 - \left(\frac{1}{1,1^3}\right)}$

$= 55403,45$

$\approx \underline{55404}$

9.3 $T_{21} = 7$

$a + 20d = 7$ (1)

$S_{10} = -550$

$\frac{10}{2}[2a + 9d] = -550$

$2a + 9d = -110$ (2)

From (1) $a = 7 - 20d$ into (2)

$2(7 - 20d) + 9d = -110$

$14 - 40d + 9d = -110$

$-31d = -124$

$\underline{d = 4}$

$a = 7 - 20(4)$

$= \underline{-73}$

$\underline{T_1 = -73}$

9.4.1. $(x^2-4)+(x^2-4)(x+3)+(x^2-4)(x+3)^2+\dots$

$$r = \frac{T_2}{T_1}$$

$$= \frac{(x^2-4)(x+3)}{(x^2-4)}$$

$$= x+3$$

To converge: $-1 < r < 1$

$$-1 < x+3 < 1$$

$$\underline{-4 < x < -2}$$

9.4.2 $$S_\infty = \frac{a}{1-r}$$

$$= \frac{x^2-4}{1-(x+3)}$$

$$= \frac{x^2-4}{1-x-3}$$

$$= \frac{x^2-4}{-x-2}$$

$$= \frac{(x-2)(x+2)}{-(x+2)}$$

$$= -(x-2)$$

$$= \underline{2-x}$$

Question 10:

10.1. $\sum_{k=1}^{\infty} 27p^{k}$

$= 27p + 27p^2 + 27p^3 + \ldots$

$S_\infty = \dfrac{27p}{1-p}$

$\sum_{t=1}^{12} (24-3t)$

$= 21 + 18 + \ldots + (-12)$

$S_{12} = \frac{n}{2}[a+l]$

$S_{12} = \frac{12}{2}[21-12]$

$= 6(9)$

$= \underline{54}$

$\therefore \dfrac{27p}{1-p} = 54$

$27p = 54 - 54p$

$81p = 54$

$p = \underline{2/3}$

10.2. $\sum_{k=1}^{\infty} 4(3-x)^k$

$= 4(3-x) + 4(3-x)^2 + 4(3-x)^3 + \ldots$

$r = \frac{T_2}{T_1}$

$= \frac{4(3-x)^2}{4(3-x)}$

$= 3-x$

To converge $-1 < r < 1$

$-1 < 3-x < 1$

$-4 < -x < -2$

$\underline{2 < x < 4}$

10.3. Areas: 4π ; 9π ; $\frac{81}{4}\pi$; ...

r: 2 ; 3 ; $\frac{9}{2}$

d: 4 ; 6 ; 9

$S_n = \frac{a(r^n - 1)}{r-1}$

$\frac{4\left(\frac{3}{2}^n - 1\right)}{\frac{3}{2} - 1} = \frac{665}{8}$

$\frac{4\left(\frac{3}{2}^n - 1\right)}{\frac{1}{2}} = \frac{665}{8}$

$4\left(\frac{3}{2}^n - 1\right) = \frac{665}{16}$

$\left(\frac{3}{2}^n - 1\right) = \frac{665}{64}$

$\left(\frac{3}{2}\right)^n = \frac{729}{64}$

$$\left(\frac{3}{2}\right)^n = \left(\frac{3}{2}\right)^6$$

$$\therefore \quad \underline{n = 6}$$

Question 11:

11.1 $\frac{a+b}{a-b} = \frac{b+c}{b-c}$

$(a+b)(b-c) = (a-b)(b+c)$

$ab + b^2 - ac - bc = ab - b^2 + ac - bc$

$2b^2 = 2ac$

$b^2 = ac$

$b = \frac{ac}{b}$

$\frac{b}{a} = \frac{c}{b}$

$\underline{\frac{T_2}{T_1} = \frac{T_3}{T_2}}$

$a; b; c$ is a geometric sequence.

11.2. $T_8 = 2\, T_4$

$a + 7d = 2(a + 3d)$

$a + 7d = 2a + 6d$

$\underline{a = d}$ ①

$S_3 = 12$

$\frac{3}{2}[2a + 2d] = 12$

$3a + 3d = 12$

$\underline{a + d = 4}$ ②

① into ②

$2d = 4$

$\underline{d = 2}$

$\therefore \underline{a = 2}$

$\underline{T_1 = 2}$

11.3. $T_1 = S_1$

$= 5 - 11$

$= \underline{-6}$

$T_2 = S_2 - S_1$

$= -2 - (-6)$

$= -2 + 6$

$= \underline{4}$

$d = T_2 - T_1$

$= 4 - (-6)$

$= \underline{10}$

11.4.1. $r = k - \frac{3}{2}$

to converge $-1 < r < 1$

$-1 < k - \frac{3}{2} < 1$

$\underline{\frac{1}{2} < k < \frac{5}{2}}$

11.4.2 $S_\infty = \frac{a}{1-r}$

$= \frac{k - \frac{3}{2}}{1 - (k - \frac{3}{2})}$

$= \frac{k - \frac{3}{2}}{\frac{5}{2} - k}$

$= \frac{\cancel{\frac{1}{2}}(2k-3)}{\cancel{\frac{1}{2}}(5-2k)}$

$= \underline{\frac{2k-3}{5-2k}}$

11.4.3(a) $\frac{1}{2} < 1 < \frac{5}{2}$ series converges

$$S_\infty = \frac{2(1) - 3}{5 - 2(1)}$$

$$= \frac{-1}{3}$$

b) -1 is not between $\frac{1}{2}$ and $\frac{5}{2}$

series does not converge.

11.5 79; 158; 237;

$a = 79$

$n = 28$

$d = 79$

$$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$$

$$S_{28} = \frac{28}{2}\left[2(79) + 27(79)\right]$$

$$= 14\ (2291)$$

$$= 32\,074 \text{ m}$$

Total walked $= 2 \times 32\,074$

$= 64\,148$ m

$= 64{,}148$ km

$\approx$ 64 km